Table 1. Statistics of human preference datasets, including Anthropic HH (Bai et al., 2022), OpenAssistant Conversations (Köpf et al., 2023), and Chatbot Arena (as of 2024/1/21). The tokens are counted by Llama2's tokenizer. "Conv" = Conversation. "Lang" = Language.

Dataset	# Convs	# Models	# Users	# Langs	Avg. # Turns per Sample	Avg. # Tokens per Prompt	Avg. # Tokens per Response
Anthropic HH	338,704	-	143	1	2.3	18.9	78.9
OpenAssistant	66,497	-	13,500	35	-	36.9	214.2
Chatbot Arena (20240121)	243,329	50	90,051	149	1.3	94.9	269.0

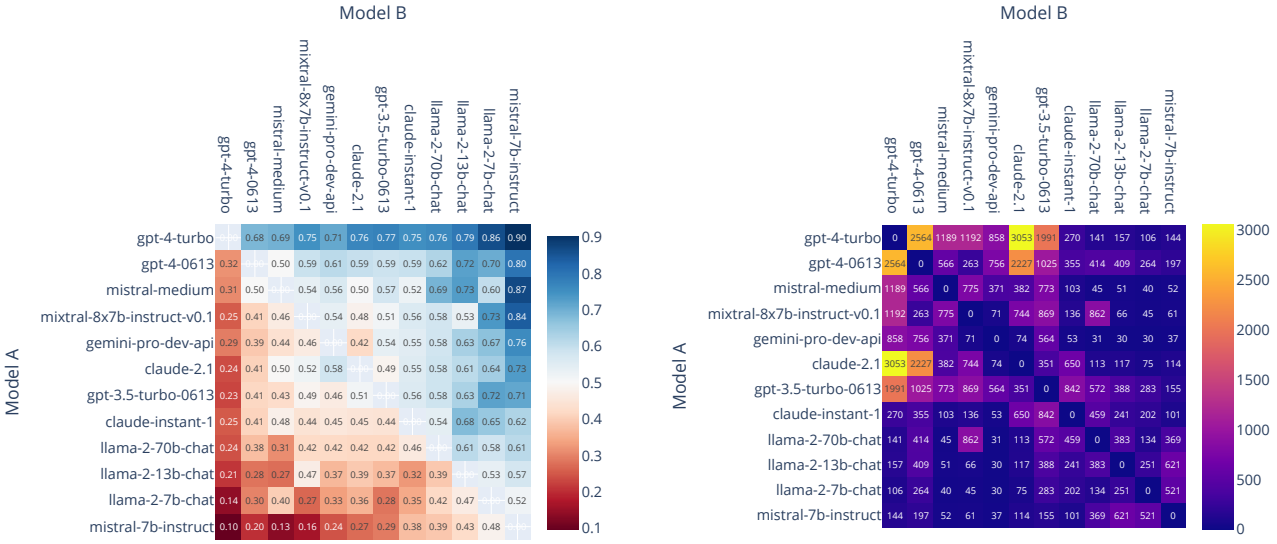

Figure 2. Win-rate (left) and battle count (right) between a subset of models in Chatbot Arena.

any form of feedback, including the possibility of allowing the human to express different degrees of preference or to say the models are tied.

One critical goal is to estimate the *win matrix*: $\theta^*(a) = \mathbb{E}[H_t \mid A_t = a]$, for all $a \in \mathcal{A}$; see the left panel of Figure 2 for an illustration of the (empirical) win matrix. In the binary case, the a entry in the win matrix corresponds to the probability the human prefers model a_2 to a_1 when shown the pair a. Finding the win matrix is a relatively straightforward mean-estimation problem; we will provide details in Section 5.

Formally, consider a *score* $s(\mathbb{P}) \in \mathbb{R}^M$, where $\mathbb{P}$ is a joint distribution over A and H (by default, we will target a uniform distribution over $\mathcal{A}$). Each model has a true score $s(\mathbb{P})_m$, and better models will have higher scores. In particular, we have the rank of model m:

$$\text{rank}(\mathbb{P})_m = 1 + \sum_{m' \in [M]} \mathbb{1}\left\{s(\mathbb{P})_{m'} > s(\mathbb{P})_m\right\}. \quad (1)$$

The best model has rank 1. If there is another model tied for best, they will both get assigned rank 1.

Picking a score. A standard score function in this setting is the vector of Bradley-Terry (BT) coefficients (Bradley & Terry, 1952). In the Bradley-Terry model, $H_t \in \{0, 1\}$, and the probability model m beats model m' is modeled via a logistic relationship:

$$\mathbb{P}(H_t = 1) = \frac{1}{1 + e^{\xi_{m'} - \xi_m}}, \quad (2)$$

where ξ is an M-length vector of so-called BT coefficients. Without loss of generality, we take $\xi_1 = 0$ (since the model is invariant to addition in ξ). Our goal is to estimate the population Bradley-Terry coefficients, i.e., those that minimize the binary cross-entropy:

$$s(\mathbb{P}) = \underset{\xi}{\text{argmin}}\, \mathbb{E}_{(A,H)\sim\mathbb{P}} \left[\ell\left(H, \frac{1}{1 + e^{\xi_{A_2} - \xi_{A_1}}}\right)\right], \quad (3)$$

where ℓ is the binary cross-entropy loss, $\ell(h, p) = -(h \log(p) + (1 - h) \log(1 - p))$.

Although the BT model technically assumes a parametric form for the model win rates, the seminal results of Huber et al. (1967); White (1982) show that maximum likelihood estimators are still asymptotically normal even when these assumptions *do not* hold, so long as the so-called "sandwich" covariance matrix is used; see Section 5 for details, and see Appendix B for a nonparametric extension of the Bradley-Terry model. Finally, we remark that previous evolutions of our online interface have reported different ranking scores, such as the Elo score (Elo, 1967) instead of the BT coefficients. We made this change because the BT coefficients are better for the purpose of statistical estimation.

5. Efficient Approximate Ranking

In Section 4 we described how to calculate the win matrix, score, and rank. Now we describe our estimation procedures.

Win matrix estimation. Estimation of the win matrix is relatively straightforward. Define $X_t(a) = \frac{1}{P_t(a)} H_t \mathbb{1}\{A_t = a\}$, where $P_t(a)$ is the probability of sampling pair a at time t, and X_t as the according vector. Then the estimator is

$$\hat{\theta}_T = \frac{1}{T} \sum_{t=1}^{T} X_t. \tag{4}$$

Note that $\mathbb{E}[X_t(a)] = \theta^*(a)$ for all t, and thus $\hat{\theta}_T$ is an unbiased estimator of θ^*. We will furthermore estimate the covariance matrix as

$$\widehat{\Sigma}_T = \frac{1}{T} \sum_{t=1}^{T} (X_t - \hat{\theta}_T)(X_t - \hat{\theta}_T)^\top. \tag{5}$$

Under the appropriate regularity conditions, we have that

$$\sqrt{T} \widehat{\Sigma}^{-1/2} (\hat{\theta} - \theta^*) \to \mathcal{N}(0, I_d), \tag{6}$$

and we construct confidence intervals accordingly. For an understanding of the appropriate regularity conditions, see Durrett (2019), Theorem 8.2.8, where condition (ii) is trivially satisfied so long as $P_t(a) > \epsilon > 0$, and condition (i) is implied by the almost-sure convergence of $P_t(a)$ to a limiting distribution $P(a)$.

Estimating the BT scores. To estimate the BT coefficients, mirroring (3), we perform (reweighted) maximum likelihood estimation on our data points:

$$s(\hat{\mathbb{P}}) = \operatorname*{argmin}_{\xi} \sum_{t=1}^{T} \frac{1}{P(A_t)} \ell\left(H_t, \frac{1}{1 + e^{\xi_{A_{t,2}} - \xi_{A_{t,1}}}}\right), \tag{7}$$

where $A_t \sim P$. We perform the inverse weighting by $P(A_t)$ because this allows us to target a score with a uniform distribution over A.

To compute confidence intervals on the BT coefficients, we employ two strategies: (1) the pivot bootstrap (DiCiccio & Efron, 1996), and (2) the "sandwich" robust standard errors outlined in Huber et al. (1967) (see also Freedman (2006) for an outline of the necessary technical assumptions). Ultimately, based on the results of a simulation study described in Appendix A, we choose to deploy the sandwich intervals due to their smaller size in large samples.

Approximate rankings. Finally, we report an approximate ranking for each model that accounts for the uncertainty in the estimation of the score. Given an M-dimensional confidence set $\mathcal{C}$ satisfying

$$\mathbb{P}(s(\mathbb{P}) \in \mathcal{C}) \geq 1 - \alpha, \tag{8}$$

we extract an approximate ranking $R_m = 1 + \sum_{m' \in [M]} \mathbb{1}\{\inf \mathcal{C}_{m'} > \sup \mathcal{C}_m\}$. The uniform validity of $\mathcal{C}$ directly implies that $\mathbb{P}(\exists m : R_m > \operatorname{rank}(\mathbb{P})_m) \leq \alpha$—i.e., with high probability, no model's performance is understated. A guarantee on the other side—that no model's performance is overstated—is possible by interchanging the $\inf$ and $\sup$. To get the uniform confidence set, we construct the chi-squared interval implied by the central limit theorem using the sandwich estimate of the variance. In other words, we construct the interval $\{\xi : T \left\| \hat{V}^{-1/2}(\hat{\xi} - \xi) \right\| \leq \chi^2_{1-\alpha, M-1}$, where $\hat{\xi}$ is our MLE of the BT coefficients and $\hat{V}_\xi$ is the sandwich variance of the logistic regression.

Active sampling rule. Our sampling rule was to choose the model pair $a \in \mathcal{A}$ proportionally to the reduction in confidence interval size by sampling that pair:

$$P_t(a) \propto \sqrt{\frac{\hat{\Sigma}_{t,a,a}}{|\{t : A_t = a\}|}} - \sqrt{\frac{\hat{\Sigma}_{t,a,a}}{|\{t : A_t = a\}| + 1}}. \tag{9}$$

5.1. Detecting Anomalous Users

On a different note, we take a first step towards identifying anomalous IP addresses in our dataset. In a dataset of U unique IPs, we let $\mathsf{IP} = \{1, \ldots, U\}$ be the set of all IP addresses. Consider a "test" user, outside this database, who gives ratings $H'_1, \ldots, H'_n$ when presented actions $A'_1, \ldots, A'_n$. The idea of our procedure is to compare the distribution of ratings for the new user to the historical distribution of ratings for a given action. We let $\mathcal{H}_a = \{H_t : A_t = a\}$ and every time a user submits a vote, we calculate the following number:

$$p_i = \frac{1}{|\mathcal{H}_{A'_i}| + 1} \left(1 + \sum_{h \in \mathcal{H}_{A'_i}} \mathbb{1}\{h \geq H'_i\}\right). \tag{10}$$

Under the null hypothesis that $\mathcal{H}_{A'_i}$ is exchangeable with H'_i, p_i is a valid p-value (see Appendix C for a proof). Furthermore, the dependence of these p-values asymptotically is negligible.

With this p-value in hand, we can test against this null hypothesis sequentially by using Fisher's combination test (Fisher, 1928) along with a variant of the Bonferroni correction. In particular, for each user, after their jth vote, we compute $M_j = -2 \sum_{i=1}^{j} \log(p_i)$. At 5 randomly chosen values of j between 1 and 100, we identify a user as anomalous if $M_j \geq \chi^2_{2j, 1-\alpha/5}$. (The times are randomly chosen, as to avoid anomalous users strategizing to hack this p-value.) Despite the heuristic application of this procedure, it seems to work well in our small-scale tests reported in Table 5.

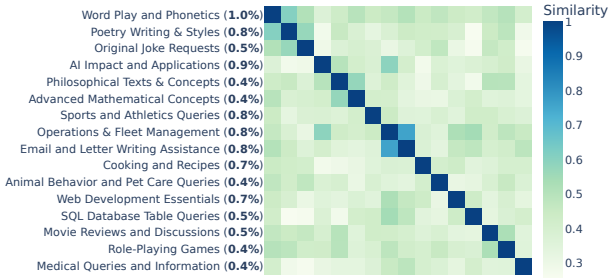

Figure 3. Similarity matrix of top-16 topic clusters. The number followed by the topic label represents the cluster size in percentage. Note that similarity is computed by cluster's centroid embeddings, hence diagonals are always one.

6. Data Analysis

To examine whether Arena's crowdsourced data reflects real-world use cases, we conduct topic modeling on the user prompts. We show how effective are these prompts in distinguishing models. Lastly, we validate the vote quality by relabeling data with experts.

6.1. Topic Modeling on User Prompts

To study the prompt diversity, we build a topic modeling pipeline with BERTopic[3] (Grootendorst, 2022). We start with transforming user prompts into representation vectors using OpenAI's text embedding model (text-embedding-3-small). To mitigate the curse of dimensionality for data clustering, we employ UMAP (Uniform Manifold Approximation and Projection) (McInnes et al., 2020) to reduce the embedding dimension from 1,536 to 5. We then use the hierarchical density-based clustering algorithm, HDBSCAN, to identify topic clusters with minimum cluster size 32. Finally, to obtain topic labels, we sample 10 prompts from each topic cluster and feed into GPT-4-Turbo for topic summarization.

The pipeline identifies 600 clusters covering a wide range of topics including poetry writing, coding, math, and medical queries. We present the top-16 topic clusters in Figure 3. We observe that the largest cluster only accounts for 1% of the entire set and the rest quickly drop to <0.5%, and the similarity between clusters is small, showing a long-tail and diverse distribution. Due to space limit, we present the similarity matrix and cluster hierarchy of top-64 clusters in Figure 11 and 12 in Appendix.

6.2. Can Arena Prompts Distinguish Models?

Next, we study how effective are these topic clusters in distinguishing models strengths. Constructing challenging prompts has become increasingly difficult due to LLMs' fast growing capabilities. For example, open models such as Llama-2-70b-chat can likely answer inquiries about movie or travel recommendation as good as GPT-4, but not in other domains such as reasoning or coding. To demonstrate, we sample 30 prompts from seven topic clusters and compare the performance of Llama-2-70b-chat and GPT-4. To control variables, we factor out user votes and consider LLM-as-judge (Zheng et al., 2023b) to evaluate model response. Results are shown in Table 2, where we see GPT-4 has significantly higher win-rate (up to 97%) in clusters that require coding and reasoning skills. On the other hand, for clusters with less problem-solving tasks, GPT-4 win-rate drops to below 60%. We show examples in Appendix D.1. This result shows models may exhibit varying strengths in different areas, but also highlights some of the topic clusters in Chatbot Arena are effective in differentiate models.

Table 2. GPT-4-0613's win-rate against Llama-2-70b-chat on 30 sample prompts from various topic clusters. We use GPT-4-turbo as judge to evaluate model responses in pairwise comparison.

Topic Cluster	Win-rate	Size
Python Game Programming Challenge	96.7%	0.2%
C/C++ Process Multi-Threading	86.7%	0.3%
SQL Query Database Assistance	73.3%	0.2%
Poetry Writing Prompts	66.7%	1.1%
Python Coding Basics	65.0%	0.2%
Linguistic Analysis & Wordplay	58.3%	0.7%
Travel Itinerary Planning	58.3%	0.4%
Movie Recommendations & Ratings	53.3%	0.2%

Building Challenging Benchmark. To further demonstrate the prompt quality, we show it is possible to construct a challenging benchmark with crowd-sourced user prompts. To ensure both topic coverage and quality, we first run the topic modeling pipeline and follow a similar procedure in Zheng et al. (2023a) to select challenging questions sampled from each topic cluster. Examples prompts and evaluation procedures can be found in the Appendix D.2 and Appendix D.3, respectively. We observe the selected prompts are highly effective in differentiating models. In Figure 4, we compare Arena bench against a widely used LLM benchmark, MT-Bench (Zheng et al., 2023b). We can see that Arena Bench effectively reveals a significant gap in performance between proprietary and the strongest open models.

6.3. Validating Vote Quality

To assess the quality of crowdsourced votes, we randomly selected 160 battles between GPT-4-Turbo and Llama-2-13B, as well as GPT-4-Turbo and GPT-3.5-Turbo-0613. We then asked experts[4] to label their preference per comparison. The experts were given the prompts and answers blindly, and asked to carefully fact-check model's answer with external resources like search engine. Manually labeling each

[3] https://github.com/MaartenGr/BERTopic

[4] The laborers are graduate students at UC Berkeley.